

Chapter 6: A Quantitative Miscellany

In our introductory chapter, we spoke of giving a friendly alien a single publication to learn about what our society and daily lives look like. There we noted that our favorite suggestion would be some report from the Bureau of Statistics that details everything from income to age at marriage. Over the last five chapters, we have performed a systematic analysis of many of the key questions that one can address as part of the main substance of the bureau of statistics of the cell: how big, how many, how forceful, how fast? Of course, there are many statistics about our society that are obscure, but still interesting, such as the number of deaths from falling, an always surprising statistic given the frighteningly high numbers and surpassing the number from food poisoning, snake bites and airplane crashes combined. Similarly, there are many interesting biological quantities that defy simple categorization, and yet, deserve mention in our pantheon of bionumbers. That is the purpose of this final chapter where we bring together some important numbers that help us understand the world of the cell and that did not fit into the categories heading the other chapters.

We now turn to a quantitative miscellany of topics that runs the gamut from exploring the characteristic state of oligomerization of the many proteins that make cells tick to the “burst size” of viruses that tell us how many new viruses will erupt from an infected cell. In each of these cases, we invite the reader to continue with the style of arguments that have been made in vignettes throughout the book and more importantly, to imagine what other interesting bionumbers would end up on their own personal quantitative miscellany. To whet the appetite for the current chapter, we thought it would be of interest to our readers to hear something more about the statistics of the searches that are made on the BioNumbers website itself. About 200 researchers every day, from across the globe find themselves curious about a very

wide spectrum of different quantities that characterize the living world. The most popular queries are independently searched for many hundred of times each year. Some of those queries fall right within the framework of our main chapters throughout the book such as how heavy is the tobacco mosaic virus, how rapid is DNA replication in humans or the microbiologist's favorite, what is the conversion from optical density units to number of cells. But there are many other search queries that do not fit at all into the framework laid out in our various chapter headings. For example, one favorite is what is the average spacing between the origins of replication on human chromosomes? Or, how many cells are in a colony? Finally, the number of hairs on a human head and the duration of the blink of an eye command great interest among internet searchers. For us the database searches show that the need for knowing the numbers that govern life is widespread and takes many forms. We hope to have given the reader a bit more of an overview of what these numbers are and how knowing them can lead to unexpected insights.

How many cells are there in an organism?

The fact that all organisms are built of basic units, namely cells, is one of the great revelations of biology. Even though often now taken as a triviality, it is one of the deepest insights in the history of biology and serves as a unifying principle in a field where diversity is the rule rather than the exception. But how many cells are there in a given organism and what controls this number and their size? The answer to these questions can vary for different individuals within a species and depends critically on the stage in life. Table 1 attempts to provide a feel for the range of different cell counts based upon both measurements and simple estimates. This will lead us to approach the classic conundrum: does a whale vary from a mouse mostly in the number of cells or is it the sizes of the cells themselves that confer these differences in overall body size?

Table 1: Number of cells in selected organisms. All values save human are based on counting using light or electron microscopy.

organism	stage in life cycle or organ	estimated cell count	BNID
human	adult	$3.7 \pm 0.8 \times 10^{13}$	102390
<i>D. melanogaster</i>	embryo cycle 14	6,000 (nuclei)	106463
<i>C. elegans</i>	adult male (somatic)	1031	100582
<i>C. elegans</i>	adult hermaphrodite (somatic)	959	100581
<i>C. elegans</i>	hatched larvae	558	101366

Perhaps the most intriguing answer to the question of cell counts is given by the case of *C. elegans*, remarkable for the fact that every individual has the same cell lineage resulting in precisely 1031 cells (BNID 100582) from one individual to the next for males and 959 cells (BNID 100581) for hermaphrodites (females also capable of self fertilization). Specific knowledge of the cell inventory in *C. elegans* makes it possible to count the number of cellular participants in every tissue type and reaches its pinnacle in the mapping of most synaptic connections among cells of the nervous system (including the worm “brain”) where every worm contains exactly 302 neurons. These surprising regularities have made the worm an unexpected leading figure in developmental biology and neuroscience. It is also possible to track down the 131 cells (BNID 101367) that are subject to programmed cell death (apoptosis) during embryonic development. Though not examined to the same level of detail, there are

other organisms besides *C. elegans* that have a constant number of cells and some reveal the same sort of stereotyped development with specific, deterministic lineages of all cells in the organism. Organisms that contain a fixed cell number are called eutelic. Examples include many but not all nematodes, as well as tardigrades (aka, water bears) and rotifers. Some of our closest invertebrate relatives, ascidians such as *Ciona*, have an apparently fixed lineage as embryos, but they do not have a fixed number of cells as adults, which arise from metamorphosis of their nearly eutelic larvae. Having a constant number of cells therefore does not seem to have any particular evolutionary origin but rather seems to be a common characteristic of rapidly developing animals with relatively small cell numbers (on the order of 1000 somatic cells).

In larger organisms, the cellular census is considerably more challenging. One route for making an estimate of the cellular census is to resort to estimates based upon volume as shown in Figure 1.

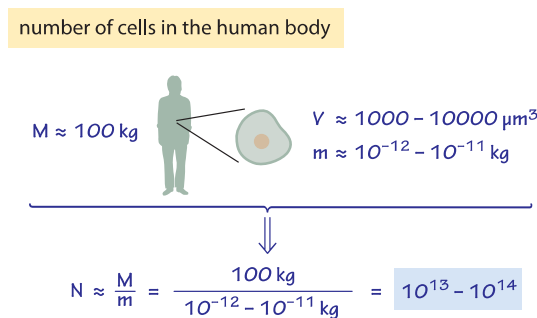


Figure 1: Estimate of the number of cells in a human body based on characteristic volumes.

For example, a human with a mass of $\approx 100 \text{ kg}$ will have a volume of $\approx 10^{-1} \text{ m}^3$. Mammalian cells are usually in the volume range $10^3 - 10^4 \mu\text{m}^3 = 10^{-15} - 10^{-14} \text{ m}^3$, implying that the number of cells is between $\approx 10^{13} - 10^{14}$ which is the range quoted in the literature (BNID 102390). Though the sizes (linear dimension) of eukaryotic organisms can vary by more than 10 orders of magnitude, the size of their cells measured by the “radius”, for example, usually varies by only a factor of ten at most except for intriguing exceptions such as the cells of the nervous system and oocytes. However, the level of accuracy of estimates like those given above on the basis of volume should be viewed with a measure of skepticism as can be easily seen by considering your recent blood test results. The normal red blood cell count is 4-6 million such cells per microliter. With about 5 liters of blood in an adult this results in an estimate of 3×10^{13} such cells rushing about in your blood stream, already for this cell type alone as many as the

total number of cells in a human body we estimated using volume arguments. The disagreement with the estimate above results from the fact that red blood cells are much smaller than the characteristic mammalian cell at about $10^2 \text{ } \mu\text{m}^3$ in volume. This shows how the above estimate should in fact be increased (and several textbooks revised). A census of the cells in the body was achieved by methodically analyzing different cell types and tissues arriving at a value of $3.7 \pm 0.8 \times 10^{13}$ cells in a human adult (BNID 109716). The breakdown by cell type for the major contributors is shown in Figure 2. The numerical dominance of red blood cells is visually clear. Of course we do not account for bacterial cells or other residents in our body, the number of cells composing this so called microbiota outnumber our human cells by a factor still unknown but probably closer to a hundred than to the often quoted value of ten.

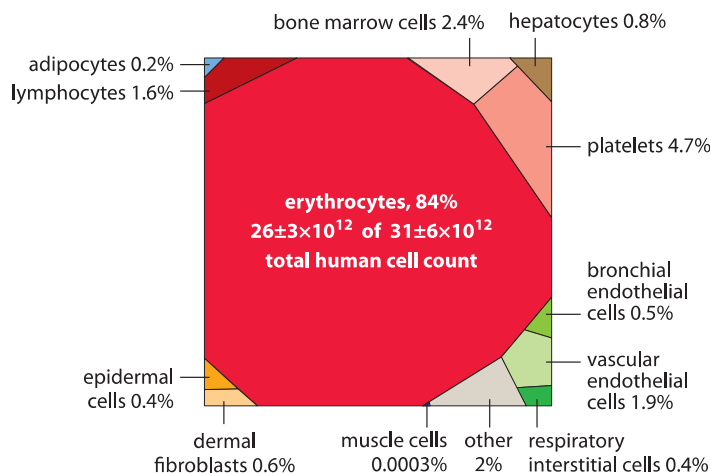


Figure 2: Estimate of the number of cells in an adult human divided by cell type. Each cell type in the human body is represented as a polygon with an area proportional to the number of cells. The dominant component is red blood cells. Based on data from R. Sender et al., in preparation, 2015.

What is the connection between organism size, cell size and cell number? Or to add some melodrama, does a whale mostly have larger cells or more cells than a mouse? In studying the large variation in fruit organ size as shown in Figure 3 it was found that the change in the number of cells is the predominant factor driving size variability. In the model plant *Arabidopsis thaliana*, early versus later leaves vary in total leaf area from 30 to 200 mm^2 (BNID 107043). This variation comes about as a result of a concomitant change in cell number from 20,000 to 130,000 with cell area remaining almost constant at 1600 μm^2 (BNID 107044). In contrast, in the green revolution that tripled yields of rice and wheat in the 1970's, a major factor was the introduction of miniature strains where the smaller

size makes it possible for the plant to support bigger grains without falling over. The smaller cultivars were achieved through breeding for less response to the plant hormones gibberellins that affects stem cell elongation. In this case, a decrease in cell size, not cell number, is the dominant factor, a change in the underlying biology of these plants that helps feed over a billion people.

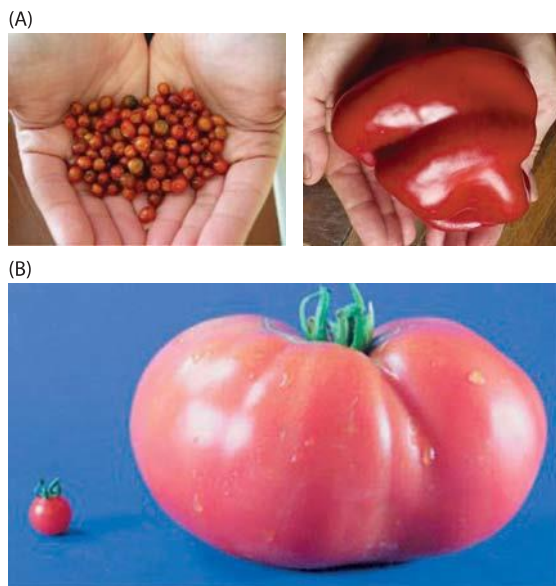
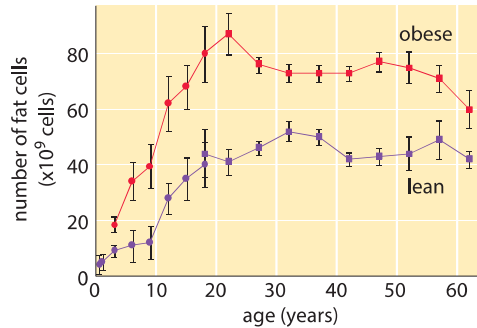


Figure 3: Plant and organ size changes from domestication, breeding hybridization and transgenic modification. These variations are found to be mostly driven by change in cell number. Fruit size of wild and domesticated species: (A) wild relative species of pepper, *Capsicum annuum* cv. Chiltepin (left) and bell pepper (right) (B) wild relative species of tomato, *Solanum* (left), *Solanum esculentum* cv Giant Red (right). (A. picture by the authors. B. Adapted from: M. Guo, C.R. Simmons / Plant Science 181 (2011) 1–7)

When the ploidy of the genome is changed the cells tend to change size accordingly. For example, cells in a tetraploid salamander are twice the size of those in a diploid salamander, although the corresponding organs in the two animals have the same size. Everything fits well because the tetraploid salamander contains half as many cells as the diploid (BNID 111481).

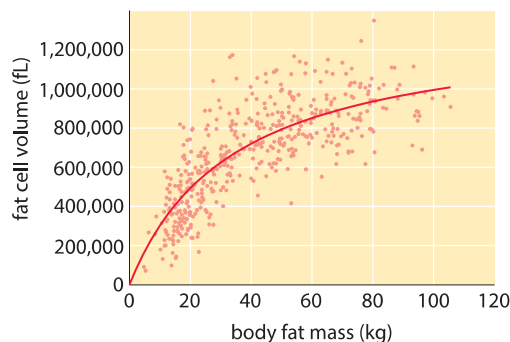
When two people differ in size, is it due to a difference in the number of cells or in the average size of cells? We can begin to answer such questions by appealing to data for lean versus obese humans. Obese adults have on the average almost twice as many fat cells. This difference between lean and obese human adults is usually established at an early age as shown in Figure 4. What about the average cell size? Figure 5 shows the variation in the average volume of a fat cell as a function of the body fat mass. At

Figure 4: Adipocyte number remains stable in adulthood, although significant weight loss can result in a decrease in adipocyte volume. Total adipocyte number from adult individuals (squares) was combined with previous results for children and adolescents (circles). The adipocyte number increases in childhood and adolescence. Lean is defined as having a body mass index <25 and obese is >30 . (Adapted from K. L. Spalding et al., Nature 453:783, 2008.)



low body fat masses the close to linear increase, passing through the origin, indicates that in this regime, differences are mostly driven by a change in the volume of the cells, i.e. the total number of cells remains relatively constant. At the high body fat range the cell volume increase is sub linear, indicating that an increase in the number of cells is becoming important. The extra fat weight, that in obese individuals can reach 100 kg, is accompanied by a change in the number of fat cells as shown in Figure 4 of less than 10^{11} which is much less than 1% of the total number of cells in the body estimated above. Thus we conclude that between lean and obese people the main change is a change in fat cell volume rather than total number of cells in the body. This is contrasted by what happens when comparing across organisms of very different sizes, say between a human and a mouse. Both organisms have cells that are usually of similar size though a person weighs more than a thousand times more. Thus in this case, and we claim this is often the case across multicellular organisms that are orders of magnitude apart, the number of cells is the main driver of size differences. With elephants having red blood cells (BNID 109091), as well as other cells, of sizes not unlike ours (BNID 109094) we hypothesize that this is also true for them.

Figure 5: Average fat cell size as a function of body fat mass. As the fat content of a person increases the average adipocyte volume initially increases almost linearly and then saturates. Thus the change in total fat among humans can be attributed mostly to larger cells of a similar number and at more extreme disparities also to change in the number of fat cells. (Adapted from K. L. Spalding et al., Nature 453:783, 2008.)



How many chromosome replications occur per generation?

A look in the mirror tells us that we are made of many different types of cells. In terms of genetics and evolution, the most important distinction is between the cells of the germ line, those cells that have the potential to culminate in new offspring, and somatic cells, those cells that build the rest of the body but do not propagate to the next generation. This dichotomy is more relaxed in plants but in principle is similar. An important factor contributing to the fidelity with which the genetic information is transferred from one generation to the next is the number of cell divisions each germ cell will make on average before the actual fertilization event.

In a previous vignette on the number of cells in an organism we made the estimate that a human is made up of 3×10^{13} cells (BNID 109716, 100290). But cells are constantly born and dying. Given this turnover, how many cells does a person make in a lifetime? Though the question touches our very own composition, we could only find one passing mention of about 10^{16} cell divisions in total during a human lifespan (BNID 100379). Our sanity check on this value relies on knowing that red blood cells are the dominant cell type by sheer number in the body (bacteria aside) and that their lifetime is on the order of 100 days. So in 100 years of life there will be about 300 cycles of replacement for these red blood cells, and the inferred total number of cell divisions is indeed of order 10^{16} ($\approx 2 \times 10^{13}$ rbc cells/person $\times$ 300 cycles in lifetime). We proceed to analyze two very naïve and extreme models regarding how many replications of the chromosomes are required to obtain the somatic cells that lead to the next generation. In the first simplified model we assume, as depicted in Figure 1, that all cells divide in a symmetric manner like a binary bifurcating tree. The number of cells progresses in a geometrical series starting from 1 at the first generation to 2, 4, 8, 16 etc. We will thus have 2^n cells after n replication rounds. 10^{13} cells will be reached after $\log_2(10^{13}) \approx 40$ replication rounds and 10^{16} cells after ≈ 50 replication rounds.

In our second toy model, we imagine an idealized process, schematically drawn in Figure 1, in which every cell in the body is a direct descendant of some single “stem cell”. In this case, the generation of the above mentioned full complement of a lifetime of cells in the human body would require 10^{16} replication rounds (maybe minus 1 to be accurate...) for the lifetime repository of cells.. Such a cell lineage model would place enormous demands on the fidelity of the replication process because mutations would accumulate as discussed in the vignette on “What is the mutation rate during genome replication?”.

The cell lineage from egg to adult is closer to the tree of binary divisions described in the first model. Nowhere has this been illustrated more dramatically than in the stunning experiments carried out to map the fates and histories of every cell in the nematode *C. elegans*. In a Herculean effort in the 1970s Sir John Sulston and coworkers delineated the full tree for *C. elegans*, redrawn in Figure 2, by careful microscope observations of the development of this transparent nematode. We can see that for this remarkable organism with its extremely conserved developmental strategy, the depth of this tree from egg to egg is ≈ 9 replications (BNID 105572).

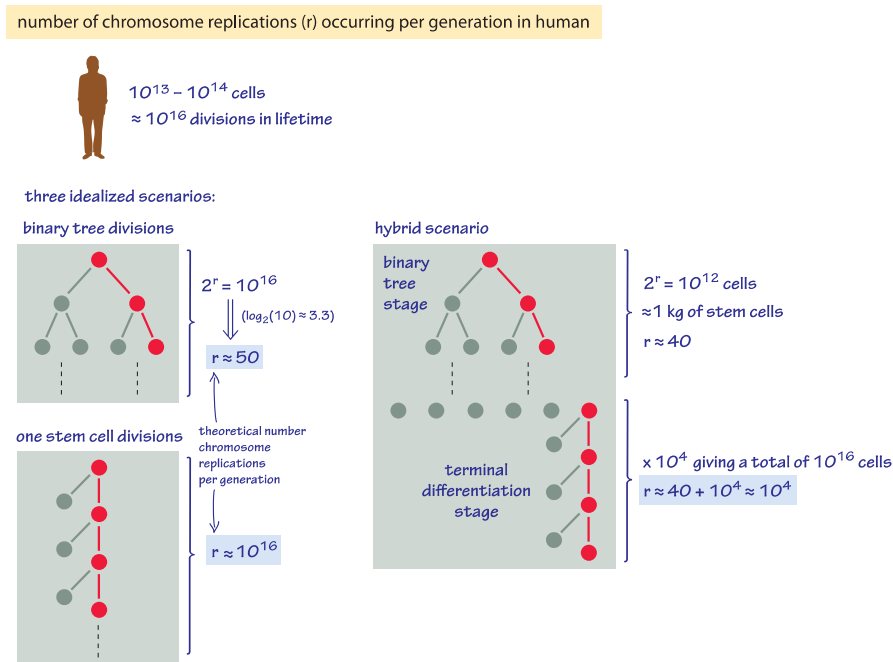


Figure 1: Back of the envelope calculation on how many chromosome replications occur per generation in simplified scenarios. For the human case the average number of divisions a cell goes through in a lifetime is different by the astronomical ratio of 50 to 10^{16} in two extreme scenarios, one based on binary tree divisions and one based on a single stem cell. A hybrid scenario starts with a binary expansion stage and transitions to a terminal division and differentiation stage.

In larger multicellular organisms, the picture is very complex and can be thought of as a hybrid between the two simplified models as also noted in Figure 1, starting from a binary tree expansion stage that then turns into a terminal differentiation stage. Such complex structured models were observed in the crypts of the colon and in the apical meristem of plants.

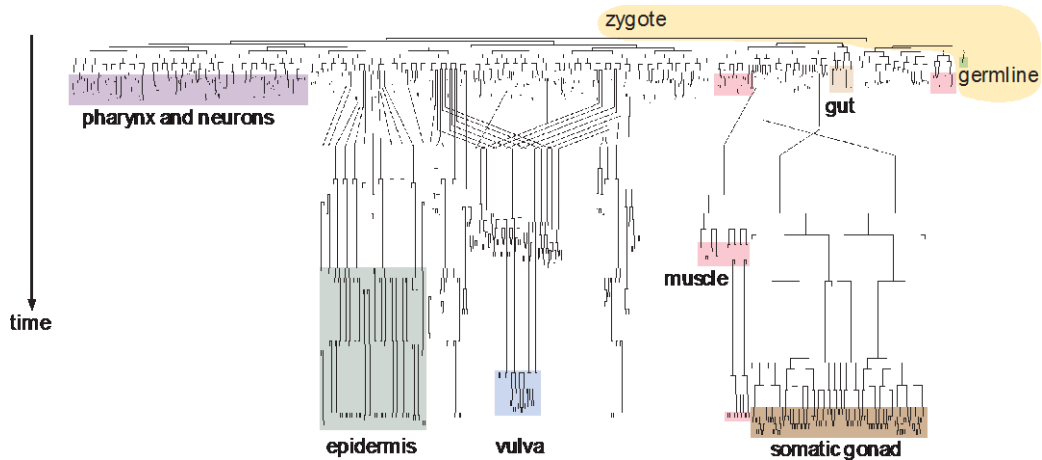


Figure 2: *C. elegans* lineage tree as deciphered through light microscopy. The path to the germline, not showing all divisions and germ cells, is highlighted in yellow.

In plants for example, a small set of stem cells, say about 10 in a plant apical meristem, divide slowly in what is termed the quiescent center. These stem cells lead to a larger population of say 100 cells that divide rapidly to give the majority of cells. The rapidly dividing cells, which accumulate mutations, are slowly replaced before they accumulate too many mutations by the progeny of the slowly dividing stem cells.

Estimates of the average number of replication rounds leading to adult cells in a range of organisms are given in Table 1. Normal mammalian cells that are not stem cells or cancerous cell lines usually stop dividing after about 40-60 cell cycles (BNID 105586). In humans, eggs are produced much earlier and in more restricted numbers than sperm cells. Indeed, there are much fewer chromosome replications from egg to egg (23, BNID 105585) versus sperm to sperm (from 35 at age 15, to >800 at age 50, BNID 105574). As mutations accumulate with replication rounds, most of the mutations arise in the male lineage. Indeed the ratio of mutations passed on by older fathers compared to mothers is about 4 to 1 (BNID 110290). Mothers are reported to pass about 15 mutations on average irrespective of age (BNID 110295) while 20 year-old fathers transmit about 25 mutations and 40 year-old fathers transmit about 65 mutations (BNID 11294). That is about 2 extra mutations per extra year of age of the father (BNID 110291). Some have gone so far as to suggest that the fact that older fathers are chiefly responsible for introducing mutations into the population, can provide a potential explanation for hemophilia in the British royal family whose kings kept on having children at advanced ages. Should the number of divisions have implications for the occurrence of cancer, which has mutations and replication at its essence? Different types of cancers are known to have very different lifetime risks that span several orders of magnitude. Recently, the number of stem cells and their division

rates are becoming available. In a recent study (C. Tomasetti & B. Vogelstein, *Nature*, 347:78, 2015), researchers collected the number of total stem cell divisions in a lifetime for 31 tissue types and correlated it to the lifetime risk of cancer occurring in that tissue. The correlation was found to be striking at about 0.8. This high correlation leaves only a much smaller fraction to be explained by environmental factors or genetic predispositions, though these have been at the center of research for decades. In our perspective this is a striking example of how paying careful attention to the numbers can still today bring simple insights into view.

Table 1: Number of chromosome replications leading to male sperm and to female ovule in different organisms

organism	number of chromosome replications leading to:		BNID
	male sperm	female ovum/ovule	
nematode <i>C. elegans</i>	8	10	105572
<i>D. melanogaster</i>	34-39	36	103523
<i>Arabidopsis thaliana</i>			106749
mouse	62	25	105576
human	40-1000	23	105574,
	[age range 15-50]		105585

How many ribosomal RNA gene copies are in the genome?

rRNA is the ribosomal RNA, a major constituent of the ribosome, accounting for about 2/3 of its mass (BNID 100119). In an earlier vignette on “How many ribosomes are in a cell?”, we discussed the large number of ribosomes required just to keep the steady pace of protein production moving. As a result of this high demand for protein production, under many growth conditions, one copy of the rRNA gene will not be enough to supply the ribosomal needs for cell growth even if that locus is being transcribed as fast as possible. Consider the budding yeast as depicted in Figure 1. Under fast exponential growth it is estimated to harbor about 200,000 ribosomes (BNID 100267). For a cell cycle time of 100 minutes, these cells need to produce ≈ 30 rRNA per second just to keep up with the demand for new ribosomes. Can the cell achieve this production rate with one gene copy? In yeast, transcription is performed by the RNA polymerase at an average speed of 10-20 bp/sec (BNID 103012, 103657). The polymerase size, or footprint over the DNA, is about 40bp (BNID 107873). Even if these genes were packed with RNA polymerase in a sequential array like cars in a traffic jam and all were moving at maximal speed, the net transcription rate would still be less than 1 rRNA/s.

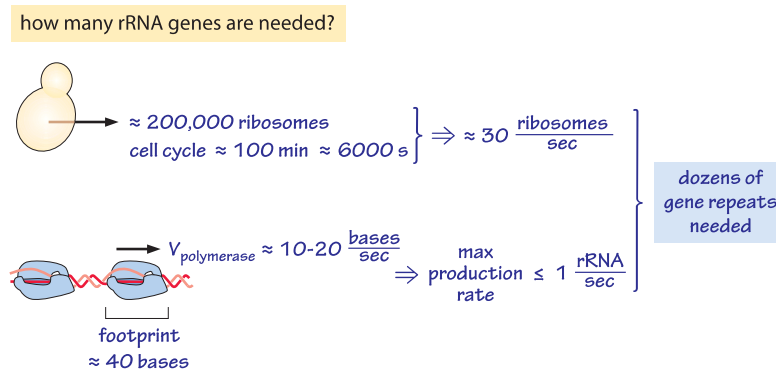


Figure 1: Back of the envelope estimation of the number of rRNA genes copies (repeats) needed in a budding yeast cell to supply the necessary ribosome production rate under fast growth conditions. Knowledge of the rough number of ribosomes and the transcription rate leads to the conclusion that several dozens of copies of the rRNA genes are needed to keep up with the demand for ribosomes.

We thus need tens of copies of the *rrna* gene to enable enough transcription to supply new ribosomes at the necessary rate. Indeed, budding yeast is known to have about 150 *rrn* (encoding the rRNA) gene copies (BNID 101733, 100243). As an aside we note how these *rrn* genes are suggested to be important players in causing aging in budding yeast (Sinclair & Guarente, Cell 91:1033, 1997).

Even in *E. coli* where very few genes appear with more than one copy per chromosome there are 7 copies of the rRNA genes as shown in Figure 2 (BNID 102219). A qualitative way to think about why there must be several copies of the rRNA gene is that rRNA is not translated so there is no second amplification step in translation as there would be for many proteins. Therefore, the only way to achieve the necessary concentration of ribosomes is to have many gene copies. The number of ribosomes in cells has been suggested to be limited by the number of operon copies with ribosomal proteins being regulated to match the synthesis rate of rRNA. The importance of the number of copies of *rrn* genes has been tested in a study in *E. coli* where *rrn* operons were deleted and the resulting growth rate was measured as a function of the *rrn* copy number for a range of 1-7 copies. With less than 6 copies there was a significant decrease in growth rate. In other experiments, extra copies beyond 7 have also been shown to be detrimental to growth, possibly because of an increase in the diffusion times as the cytoplasm becomes ever more packed with ribosomes (T. Asai et al, J Bact. 181:3803, 1999; D. A. Schneider & R. L. Gourse, J Bact. 185:6185, 2003; A. D. Tadmor & T. Tlusty, PLOS comp bio, 4:5, 2008).

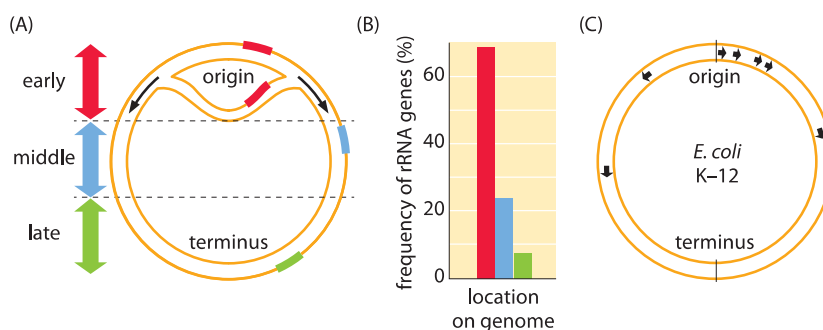


Figure 2: Ribosomal RNA genes across microbial genomes. (A). Frequency in each third of the chromosome of rDNA operons in 68 bacterial genomes. (B). Locations of the rRNA genes on the circular genome of *E. coli*. Note that all copies of the RNA genes are on the leading strand and none are on the lagging strand. (A, adapted from E. P. C. Rocha Microbiology 150:1609, 2004. B, Adapted from M Nomura PNAS;96:1820, 1999).

The number of rRNA copies in *E. coli* has apparently remained constant at 7 copies per genome over the ≈ 150 million years since it diverged from *S.*

typhimurium (BNID 107087, 107867). As *E. coli* can have nested replication forks resulting in multiple DNA replication operons close to the origin, under fast growth rates there will be more copies of the origin. Indeed, the *rrn* operons tend to be close to the origin and the number of copies per cell can be much higher than 7 – at high growth rate this number can be as high as 36 (BNID 102359). Advances in genomics allow us to retrieve the number of rRNA genes of hundreds of organisms and these numbers are now summarized in databases (BNID 104390). The databases tables show how the number of copies tends to be higher in faster growing cells. Information on their location along the genome can show for example how common is the tendency to cluster near the origin as shown in Figure 2B. One can also observe the tendency of rRNA to be coded on the leading strand (arrow direction facing away from the origin in Figure 1A) rather than on the lagging strand in replication as further discussed in the caption. This can be understood through the “collision avoidance” model. When the replication fork overtakes a transcribing RNA polymerase moving in the same direction there is replication slow down until transcription ends. When the RNA polymerase is moving in the direction opposite to the fork advance there is a head on collision that leads to a much more problematic replication arrest and transcription abortion. Avoidance of these latter cases selects for locating RNA genes on the leading strand (E. P. C. Rocha, *Ann. Rev. Genet.* 42:211, 2004).

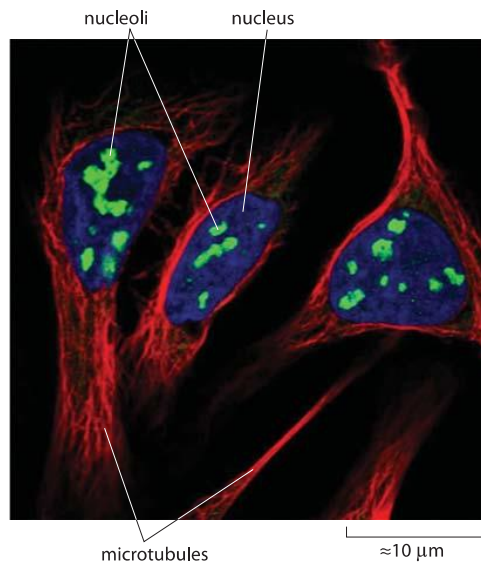


Figure 3: Visualizing the nucleolus, the location of ribosomal RNA transcription and assembly in the human cell line U-2 OS. Immunofluorescent staining for the gene RRP15 that is located in the nucleolus and functions in the maturation of the ribosomal subunit. Staining of gene (nucleoli) in green, with additional staining of nucleus in blue and microtubules in red. (Adapted from the human protein atlas project: <http://www.proteinatlas.org/ENSG00000067533/subcellular>)

How does this question of number of ribosomal RNA genes play out in the case of *Homo sapiens*? Humans carry about 200 copies of the *rrn* genes in their genome (BNID 107865), these are organized in 5 clusters known as the nucleolus organizers each containing multiple copies of the rRNA operon. In phase microscopy of human cells these areas of extensive transcriptional activity known as nucleoli are vividly seen as black dots inside the nucleus and even more strikingly in fluorescent microscopy as seen in Figure 3. Clearly, in organisms ranging from bacteria to humans, the number of ribosomes is a critical cellular parameter and one of the main ways that it is regulated is through the gene copy number itself.

What is the permeability of the cell membrane?

One of the signature characteristics of all living organisms is that they contain a distinctive mixture of ions and small molecules. The composition not only differs from the environment but can also vary within the cell. For example, the concentration of hydrogen ions in some cellular compartments can be 10^4 times greater than in others (the mitochondria reaching a pH as high as 8; the lysosomes with a pH as low as 4, BNID 107521, 106074). The ratio of the concentrations of Ca^{2+} ions in the extra- and intracellular fluid compartments can once again be 10^4 -fold (BNID 104083). This concentration difference is so large that transporting a Ca^{2+} ion across the membrane, from the intra- to the extracellular compartment, requires the energy of more than one proton or sodium ion flowing down the proton-motive force gradient. To see this, the reader should remember the rule of thumb from our tricks of the trade list that to establish an order of magnitude potential difference requires 6 kJ/mol ($\approx 2 k_{\text{B}}T$). This energy can be attained for example by transport of one electric charge through a 60 mV potential difference. To achieve four orders of magnitude concentration ratio would then require a charge to travel down about 240 mV of electron motive force (actually even more due to the double charge of the calcium ion). This is very close to the breakdown voltage of the membrane as discussed in the vignette on “What is the electric potential difference across membranes?”. Indeed the high concentration ratio of Ca^{2+} is usually achieved by coupling to the transport of three sodium ions or the hydrolysis of ATP, which helps achieve the required density difference without dangerously energizing the membrane.

The second law of thermodynamics teaches us that, in general, the presence of concentration gradients will eventually be bled off by mass transport processes, which steadily drive systems to a state of equilibrium. However, although the second law of thermodynamics tells us the nature of the ultimate state of a system (e.g. uniform concentrations), it doesn't tell us how long it will take to achieve that state. Membranes have evolved to form a very effective barrier to the spontaneous transfer of many ionic and molecular species. To estimate the time scale for equalizing concentrations, we need to know the rates of

mass transport, which depend upon key material properties such as diffusion constants and permeabilities.

A hugely successful class of “laws”, which describe the behavior of systems that have suffered some small departure from equilibrium are the linear transport laws. These laws posit a simple linear relation between the rate of transport of some quantity of interest and the associated driving force. For mass transport, there is a linear relation between the flux (i.e. the number of molecules crossing unit area per unit time) and the concentration difference (which serves as the relevant driving force). For transport across membranes, these ideas have been codified in the simple equation (for neutral solute) $j = -p^*(C_{in} - C_{out})$, where j is the net flux into the cell, C_{in} and C_{out} refer to the concentrations on the inside and outside of the membrane bound region, and p is a material parameter known as the permeability. The units of p can be deduced by noting that flux has units of number/(area x time) and the concentration has units of number/volume, implying that the units of p itself are length/time. Like many transport quantities (e.g. electrical conductivities of materials which span over 30 orders of magnitude), the permeability has a very large dynamic range as illustrated in Figure 1. As seen in the figure, lipid bilayers have a nearly 10^{10} -fold range of permeabilities.

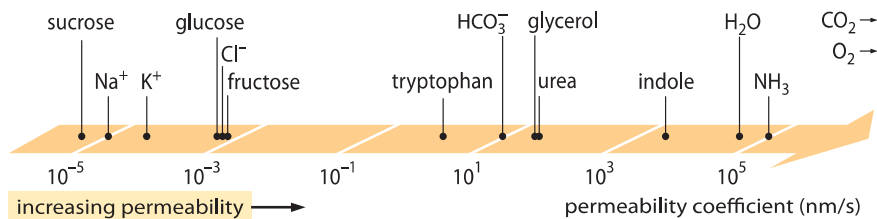


Figure 1: The wide range of membrane permeabilities of different compounds in the cell. Membranes are more permeable to uncharged compounds and least permeable to charged ions. Note that the existence of ion channels will make the apparent permeability several orders of magnitude higher when these channels are open. The units are chosen as nm/s and several nm is the characteristic membrane width. Figure adapted from R. N. Robertson, *The Lively Membranes*, Cambridge University Press, 1983. The value for glucose is smaller than in Robertson based on several sources such as BNID 110830, 110807. Other sources of data: BNID 110729, 110731, 110816, 110824, 110806.

What physico-chemical parameters guide the location of a compound on this scale of permeabilities? One rule of thumb is that small molecules have higher permeabilities than larger molecules. Another rule of thumb is that neutral compounds can cross the membrane many orders of magnitude faster than similar charged compounds. Among the charged

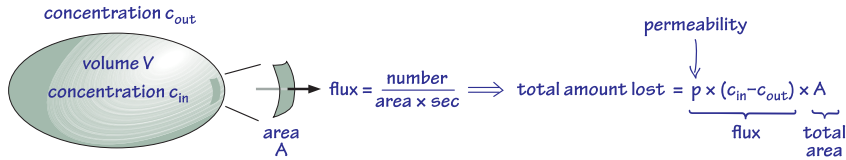
compounds, negative (anionic) compounds tend to have much higher permeabilities than positive (cationic) compounds. The so called Overton rule states that membrane permeability increases with hydrophobicity, where hydrophobicity is the tendency of a compound to prefer a non-polar solvent to a polar (aqueous) solvent. The Overton rule predicts that charged molecules (non-hydrophobic), such as ions will tend to have low permeability as they incur an energetic penalty associated with penetrating the membrane, whereas dissolved gases such as O₂ and CO₂, which are hydrophobic (as they are uncharged and symmetric), will have high permeability. Indeed, the permeability of lipid bilayer membranes to CO₂ give values that are 0.01-10 cm/sec (yes, permeability measurements have very high uncertainties among different labs, BNID 110004, 110617, 102624), higher than all other values shown in Figure 1. This value shows that the barrier created by the cell membrane is actually less of an obstacle than the barrier caused by the unstirred layer of water engulfing the cell membrane from the outside. Such an inference can be derived by the equation for the permeability coefficient of an obstacle, given by $p = K \times D/l$ where l is the width, D the diffusion coefficient and K the partition coefficient between the media and the obstacle material. This is also known as the “solubility-diffusion” model for permeability where these denote the K and the D effects which are two steps affecting the permeability. For an unstirred layer of water $K=1$ as it is very similar to the media but for membrane the value for all but the most hydrophobic material is usually several orders of magnitude smaller than 1. This dependence on K is at the heart of the Overton rule mentioned above. The high permeability for CO₂ also suggests that channels such as aquaporins that were suggested to serve for gas transport into the cell are not required as the membrane is permeable enough. To see how the membrane properties affect the chemical makeup of metabolites we turn to calculating the time of leakage for different compounds.

We consider glycerol, for example. The analysis shown in Figure 2 gives an estimate for the time of its leakage out of the cell if the molecule is not phosphorylated or otherwise converted into a more hydrophilic form. The permeability of the cell membrane to glycerol is $p \approx 10\text{-}100$ nm/s (BNID 110824) as can be read from Figure 1. The time scale for a glycerol molecule inside the cell to escape back to the surrounding medium, assuming no return flow into the cell ($c_{\text{out}}=0$), can be crudely estimated by noting that the efflux from the cell is $p \cdot A \cdot c_{\text{in}}$ where A is the cell surface area. The time scale is found by taking the total amount in the cell, $V \cdot c_{\text{in}}$ (where V is cell volume or more accurately the cell water volume), and dividing by this flux resulting for a bacterial cell ($r \approx 1 \mu\text{m}$) in a time scale:

$$t = V \cdot c_{\text{in}} / p \cdot A \cdot c_{\text{in}} = (4\pi r^3/3) / (4\pi r^2 \cdot 30 \text{ nm/s}) \approx 10 \text{ s.}$$

This is a crude estimate because we did not account for the decreasing concentration of c_{in} with time that will give a correction factor of $1/\ln(2)$, i.e less than 2 fold increase. What we learn from these estimates is that if the glycolytic intermediates glyceraldehyde or dihydroxyacetone which are very similar to glycerol were not phosphorylated, resulting in the addition of a charge, they would be lost to the medium by diffusion through the cell membrane. In lab media, where a carbon source is supplied in abundance, this is not a major issue, but in a natural environment where cells are often waiting in stationary phase for a lucky pulse of nutrients (*E. coli* is believed to go through months of no growth after its excretion from the body before it finds a new host), the cell can curb its losses by making sure metabolic intermediates are tagged with a charge that will keep them from recrossing the barrier presented by the lipid bilayer.

leakage timescale through membrane (rapid if small molecule is uncharged e.g., glycerol)



leakage timescale, τ

$$\tau = \frac{\text{number of molecules inside}}{\text{amount lost each second}} = \frac{V \times c_{in}}{p \times A \times (c_{in} - c_{out})} = \frac{c_{in} \times \frac{4}{3} \pi r^3}{c_{in} \times p \times 4 \pi r^2} = \frac{r}{3p}$$

assume $c_{out} = 0$

for glycerol in *E. coli*

$$\tau = \frac{r}{3p} \approx \frac{1}{3} \times \frac{10^{-6} \text{ m}}{4 \times 10^{-8} \frac{\text{m}}{\text{s}}} \approx 10 \text{ s}$$

bacterial cell size

glycerol permeability

so if the similar glyceraldehyde used in glycolysis was not phosphorylated it would rapidly leak from cell

Figure 2: Back of the envelope calculation of the timescale for unphosphorylated glucose molecule to passively diffuse out of a bacterial cell. The functional implications are then considered for fast growing cell where the effect is negligible and for cells in stationary state where it may cause an appreciable leakage of resources.

How many photons does it take to make a cyanobacterium?

Autotrophs are those organisms that are able to make a living without resorting to preexisting organic compounds and as such, are the primary producers of organic matter on planet Earth. One of the most amazing autotrophic lifestyles involves the use of inorganic carbon in the form of CO_2 and the synthesis of organic carbons using light as the energy input, the phenomenon known as photosynthesis. Chemoautotrophs carry out a similar performance, though in their case, the energy source is not light from the sun, but some other terrestrial energy source such as a thermal vent in the ocean or a reduced inorganic compound such as molecular hydrogen or ferrous iron.

Photoautotrophs refer to the sum total of those organisms that take energy from sunlight and convert it into organic compounds that can be oxidized. The most familiar examples are the plants that surround us in our forests and gardens. However, the overall synthetic budget goes well beyond that coming from plants and includes algae and a variety of microscopic organisms including single-celled eukaryotes (protists) and a whole range of prokaryotes such as cyanobacteria (formerly known as blue-green algae).

The majority of the Earth's surface is covered by water and photosynthesis in these great aqueous reservoirs is a significant fraction of the total photosynthetic output across the planet as a whole. Aquatic photosynthesis is largely performed by organisms so small they are not visible to the naked eye. Despite their macroscopic invisibility, these organisms are responsible for fixing ≈ 50 gigatons (BNID 102936, 10^{39} CO_2 molecules) of carbon every year. This accounts for about one half of the total primary productivity on earth (BNID 102937) but the vast majority of this fixed carbon is soon returned to the atmosphere following rapid viral attacks, planktonic grazing and respiration (BNID 102947). The process of transforming inorganic carbon into the building blocks of the organic world occurs through the process of carbon fixation, where the energy from about 10 photons is used in order to convert CO_2 into a carbohydrate, $(\text{CH}_2\text{O})_n$. The H is donated by water as an electron donor, which is thus transformed into oxygen. By comparing the combustion energy stored per carbon in carbohydrates versus the energy in solar flux leading to 10 photosynthetically active photons the overall theoretical

conversion efficiency can be calculated to be about 10%. This is similar to current off the shelf photovoltaic cells that result in electricity rather than carbohydrates as output. The biological efficiency that is actually realized is usually lower by another 1-2 orders of magnitude due to respiration, light saturation and other processes but on the other hand these photosynthetic machines can reproduce and heal themselves which cannot be said of silicon cells.

The process of oxygenic photosynthesis was invented about 3 billion years ago and transformed our atmosphere from one with practically no oxygen to one where abundant oxygen allows the existence of animals like us. Much of the carbon fixation happens in small organelles like those shown in Figure 1 and known as carboxysomes. Carboxysomes exist in some photosynthetic prokaryotes and are home to an army of molecules which perform the carbon fixation process through a key carboxylating enzyme, Rubisco, considered by many to be the most abundant protein in the biosphere. To make its ubiquity more tangible, if we were to distribute it from autotrophs to humans there would be about 5 kg of this protein per person on earth (BNID 103827).

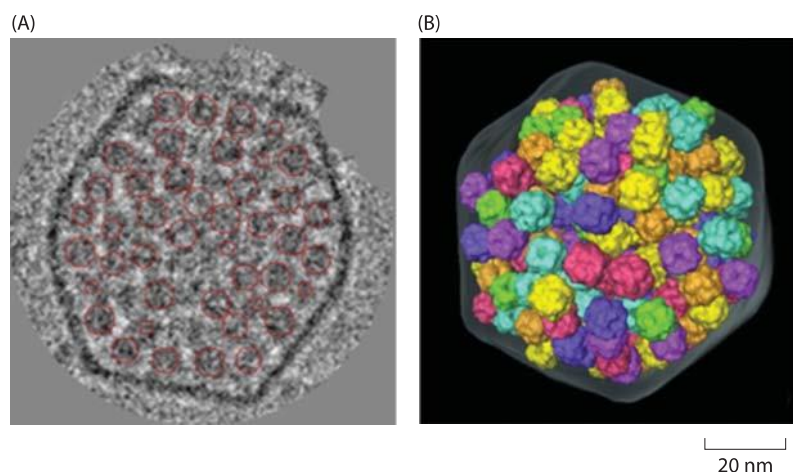


Figure 1: The structure of carboxysomes and the Rubisco octamers occupying them as determined using cryo electron microscopy. The sizes of individual carboxysomes in this organism (*Synechococcus* strain WH8102) varied from 114 nm to 137 nm, and were approximately icosahedral. There are on average ≈ 250 Rubisco octamers per carboxysome, organized into three to four concentric layers. *Synechococcus* cells usually contain about 5-10 carboxysomes. (Adapted from C. V. Iancu et al., *Journal of Molecular Biology*, 372:764, 2007.)

From an order of magnitude perspective, it suffices to think of a cyanobacterium as similar in chemical composition to a conventional bacterium, which means that it takes roughly 10^{10} carbons (see vignette on “What is the elemental composition of a cell?”) to supply the building materials for a new cyanobacterium with a volume of $1 \mu\text{m}^3$ as depicted in figure 2. Given that it requires roughly ten photons to fix a carbon atom, this implies roughly 10^{11} photons are absorbed to fix those 10^{10} carbons. This carbon fixation is carried out by roughly 10^4 Rubisco monomers within a given cyanobacterium. What about the energy required for other cellular processes such as amino acid polymerization into proteins and keeping the membrane potential maintained to drive a multitude of coupled reactions? For bacteria these energetic requirements were estimated to be on the order of 10^{10} ATP as discussed in the vignette on “What is the power consumption of a cell?”. Given that a photon can be used to produce more than one ATP equivalent (through extruding protons or electron storage in NADPH) we find that the burden of carbon fixation is dominant over these other biosynthetic and maintenance tasks.

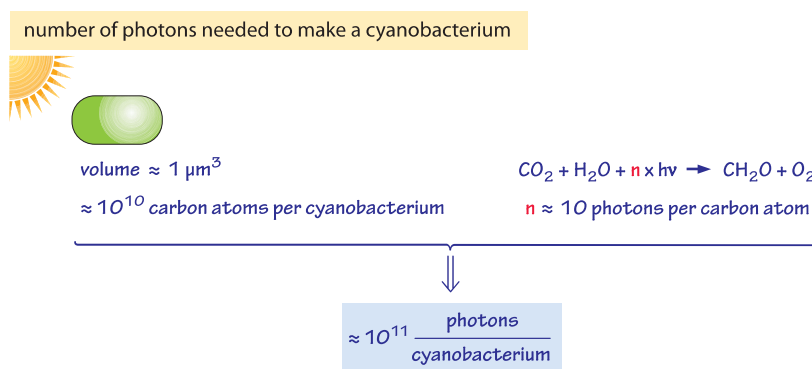


Figure 2: Order of magnitude estimation of the number of photons required to build a cyanobacterium.

How many virions result from a single viral infection?

Viruses proliferate in natural environments by infecting cells and hijacking their replication and protein synthesis machinery. After new viral proteins are synthesized and assembled, bursts of viruses are released from the infected (and usually soon to be dead) cells to repeat the process all over again. How many viruses are released from each infected cell? This parameter is referred to as the viral burst size, alluding to the fact that often virus emission either leads to cell lysis (bacteriophage) or cell death (HIV infection of T-cells). The emission of new viruses from an infected cell hence occurs as a burst with characteristic numbers of viruses and with time scales lasting from minutes to days depending upon the kind of virus and host. Burst sizes for different viruses have a large range corresponding in turn with the range of different sizes of the host cells. For example, SIV, a cousin and model for the HIV virus, is released from infected T cells with a burst size of $\approx 50,000$ (BNID 102377) whereas cyanobacterial viruses have characteristic burst sizes of ≈ 40 -80 (BNIDs 103247, 104841, 104842) and phage lambda and other phages (such as T4, T5 and T7) attacking bacteria have burst sizes of ≈ 100 -300. (BNID 105025, 105870). An example of a host bacterium prior to the burst process itself is shown in Figure 1.

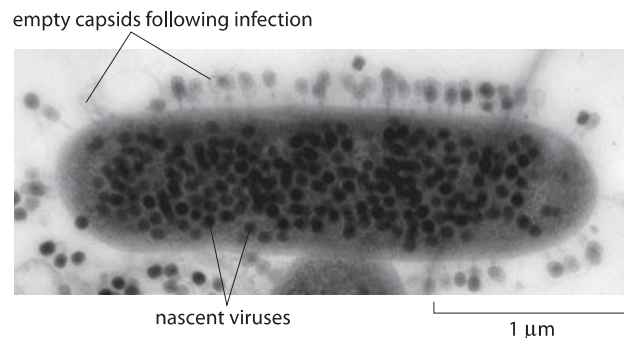


Figure 1: A transmission electron micrograph of a thin section of *Escherichia coli* K-12 infected with Bacteriophage T4. Dark viruses on the outside are ones that did not eject their DNA into the bacterial host. Image courtesy of John Wertz.

One interesting way to garner an impression of the impact of a viral infection on the host metabolism is by thinking about the volume taken up by the newly synthesized viruses in comparison with the size of the host cell. In particular, we ask what fraction of the host cell volume is occupied by all the viruses making up a viral burst? These volumes can be thought of as a proxy for biomass and thus reflect on the cell's resources that were seized. An SIV virion is roughly 100 nm in diameter and the host cell has a corresponding diameter of about 10 μm . Given these numbers, 50,000 virions thus represent about 5% of the cell's volume as shown schematically in Figure 2. In the case of bacteria and the viruses that infect them, a T-phage with ≈ 50 nm diameter (BNID 105870) shows burst sizes of ≈ 200 in an *E. coli* cell, representing $\approx 2\%$ of the volume. Therefore, the characteristic volume fraction taken up by the viruses in these two very distinct cell types shows a much smaller range (<3 fold) than the absolute burst sizes range (>100 fold). This may reflect limits to how much biomass viruses can extract from infected cells. In the marine environment which is often depleted in phosphorus that is mostly required for nucleotides, it was suggested that the DNA sizes of the virus and host govern the virion burst size as the virus utilizes the host DNA building blocks (C. M. Brown *et al.*, J. Mar. Bio. Ass. U.K., 86:491, 2006). The measurements and estimates throughout this vignette raise the very interesting question of what governs the overall burst size, as well as what fraction of the synthesized viral DNA and proteins actually make it into infectious viruses.

fraction of cell volume occupied by bursting virions

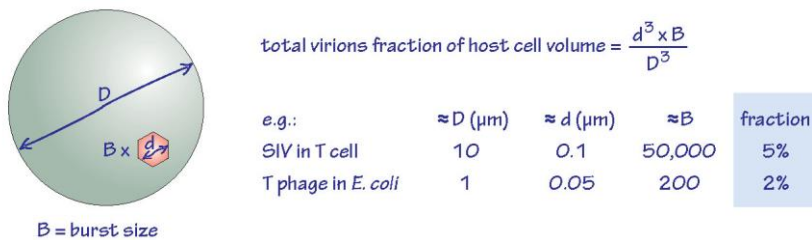


Figure 2: Back of the envelope calculation showing how the fraction of volume occupied by bursting virions is roughly similar in conditions of very different burst size. T phages capsids come in different sizes, the value chosen here is characteristic of T1, T3 and T7 (BNID 105870).

Table 1: Virons burst sizes from various host organisms. Table focuses on contrasting prokaryotes versus mammalian cells

virus	host cell used	burst size	BNID
multicellular host			
influenza A & B	chicken egg cells	500-1,000	101590
influenza A	MDCK cell line	1,000-10,000	101605
HIV	<i>H. sapiens</i> memory T cells	1,000-3,000	105872
SIV (model for HIV)	<i>R. macaque</i> T cells	40,000-60,000	102377
prokaryotic host			
cyanomyovirus S-PM2	<i>Synechococcus WH7803</i>	40	104841
podovirus P60	<i>Synechococcus WH7803</i>	80	104842
cyanomyovirus MA-LMM01	<i>M. aeruginosa</i>	50-120	103247
bacteriophage S1	<i>Stenotrophomonas sp.</i>	80	104855
bacteriophage S3	<i>Stenotrophomonas sp.</i>	100	104852
phi EF24C	<i>E. faecalis</i>	100	104857
bacteriophage Lambda	<i>E. coli</i>	150	105025
bacteriophage T1 to T7	<i>E. coli</i>	100-300	105870
bacteriophage MS2	<i>E. coli</i>	5,000-10,000	109050

Epilogue

In the course of the nearly 100 separate vignettes that make up this book, our work has been animated by several key ideas. First, the overarching theme of the book is that biological numeracy expands our view of the living world in a way that can reveal new insights into organisms and how they work that would otherwise be hidden. It can be thought of as a sixth sense complementing the already powerful arsenal of modern biology. In order to make biological numeracy useful the values reported for key biological parameters need to be characteristic and actually mean something. To that end, each of our vignettes has tried to report on carefully vetted, state-of-the-art data for a variety of key numbers that dictate the behavior of living matter. But it is not enough to merely quote the numerical values of these quantities. They must also be provided some context such that they are actually consonant with what we understand about biological systems. Hence, a second key thrust of our vignettes has been to adopt an attitude of order-of-magnitude thinking to try and use simple estimates to illuminate biological problems in a way that leaves us with an intuition for the meaning of these numbers.

Some challenges make the task of those seeking biological numeracy from reading the literature harder than one might have imagined. One challenge relates to the limited availability of numbers in textbooks and online resources and their often unclear connection to the primary literature. We hope that through efforts such as the BioNumbers database and this book we have helped remedy some of that challenge. Another challenge we have mentioned several times throughout the book are the misunderstandings that can exist when discussing absolute numbers of some cell component or other property of “the cell” without knowing the cell growth conditions. Differences in cell size can be as much as several fold and growth rate or different physiological conditions can create even further uncertainty by changing